

Kunal

Machine Learning Developer

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TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C++

Frameworks: Scikit-learn, TensorFlow, PyTorch

Libraries: Pandas, NumPy, Matplotlib, Seaborn

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Feature Selection, Model Evaluation, Hyperparameter Tuning, Cross-Validation

Generative AI: LangChain, Hugging Face Transformers, OpenAI API, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, ChromaDB)

Databases: MySQL, PostgreSQL

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, VS Code

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Probability & Statistics, Linear Algebra

PROJECTS

Image Compression Model – K-Means Clustering Approach | [GitHub](#)

2025

- Built an unsupervised image compression model using K-Means clustering to reduce image size by up to 90% with minimal perceptible quality loss.
- Implemented K-Means-based colour quantization to group similar pixel values, replacing them with representative cluster centroids for efficient pixel grouping.
- Designed the solution to be lightweight and computationally efficient, making it suitable for web platforms, storage optimization, and faster image transmission without heavy computational requirements.

Bhopal Air Quality Prediction – PM2.5 & AQI Forecasting | [GitHub](#)

2025

- Developed a machine learning model to predict PM2.5 concentration and overall AQI levels in Bhopal using historical weather and pollutant data.
- Performed data collection, cleaning, and preprocessing on multi-source weather and pollutant datasets, handling missing values and inconsistent time-series entries.
- Engineered features from meteorological parameters (temperature, humidity, wind speed) and pollutant concentrations to improve model input quality.
- Trained and evaluated regression models to forecast PM2.5/AQI trends, and visualized pollution patterns using Matplotlib/Seaborn to support interpretability.

EDUCATION

VIT Bhopal University

Sep 2023 – May 2027

B.Tech in Computer Science Engineering (Specialization in AI&ML) — **CGPA: 8.59**

Greenland Public School

Apr 2021 – Apr 2023

CBSE — Class XII: 77.4% — Class X: 95.4%

LEADERSHIP & ACTIVITIES

Founder & Business Analyst, GrabKrt – led non-technical operations (vendor onboarding, market research, outreach) for a pan-India e-commerce venture.

PR & Outreach Member, Linux Club, VIT Bhopal – organized sponsors and guest coordination for the university's flagship fest and maintained all the records or reports of the event.

Volunteer, NSS Unit, – managed crowds/donors at a blood donation camp and supported a 10,000-sapling plantation drive.

HACKATHONS & CERTIFICATIONS

Participant – HackShastra (Unstop), CyberShield Hackathon (VIT Bhopal), Sandisk Hackathon (HackerEarth).

NPTEL Cloud Computing – Elite Certificate (IIT Kharagpur)

Applied Machine Learning in Python – University of Michigan